

JASMIN

Overview for training workshops



Science and
Technology
Facilities Council



Natural
Environment
Research Council

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April 2026

History

- Early 2012
 - CEDA Archive of environmental data was outgrowing its hosting environment
 - Increasing need to support scientific workflows for analysing data, as well as “just” downloading data
 - To tackle this, JASMIN was created
- Design considerations
 - Avoid the need to move large volumes of data around
 - Enable collaboration around data
 - Provide appropriate balance between flexibility & performance



Organisation

- Funded primarily by NERC to support the environmental science community
- Infrastructure operated by STFC Scientific Computing
- User-facing services run by CEDA

Result:

- Hosting environment for CEDA Archive & related services
- Co-located HPC data analysis platform
- Support from specialist teams



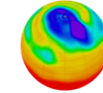
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Scientific Computing

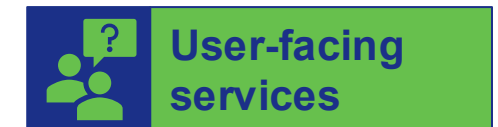


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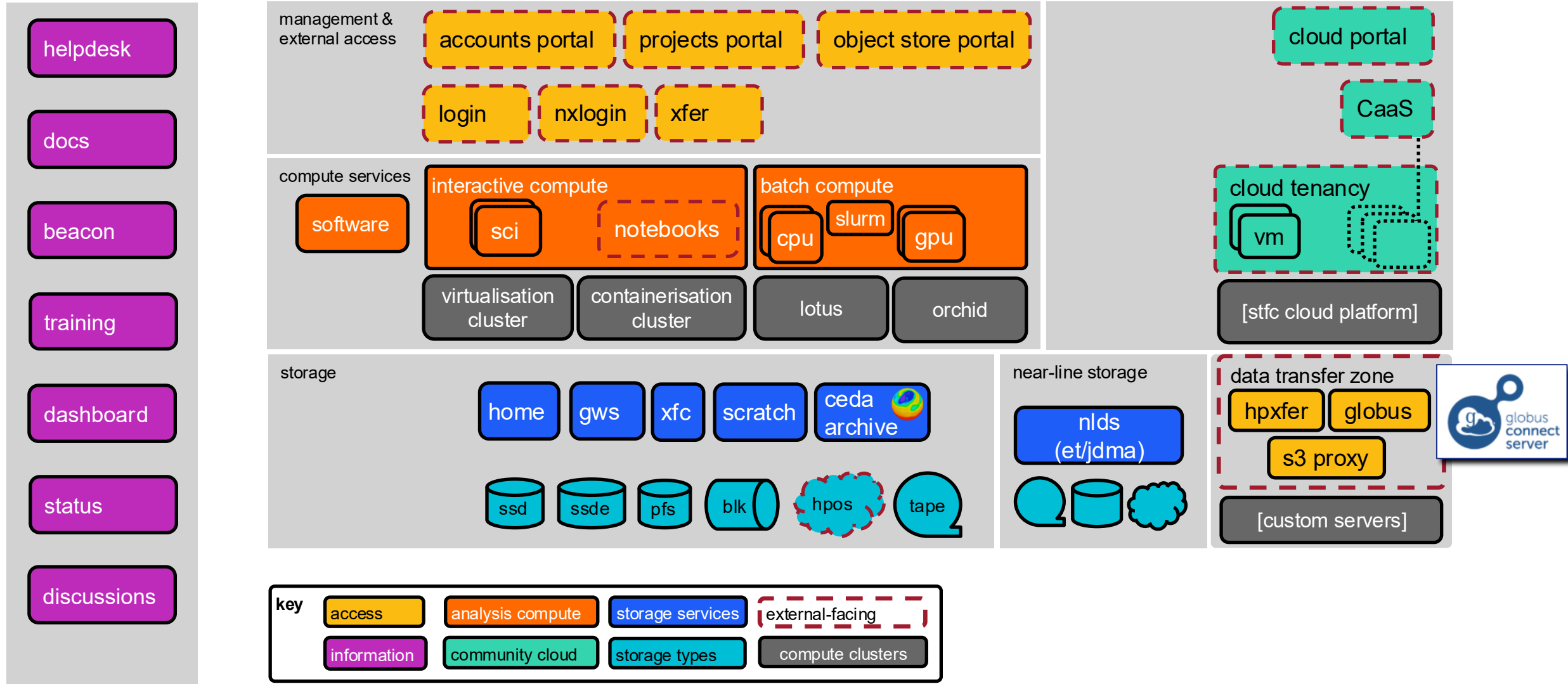
Natural
Environment
Research Council

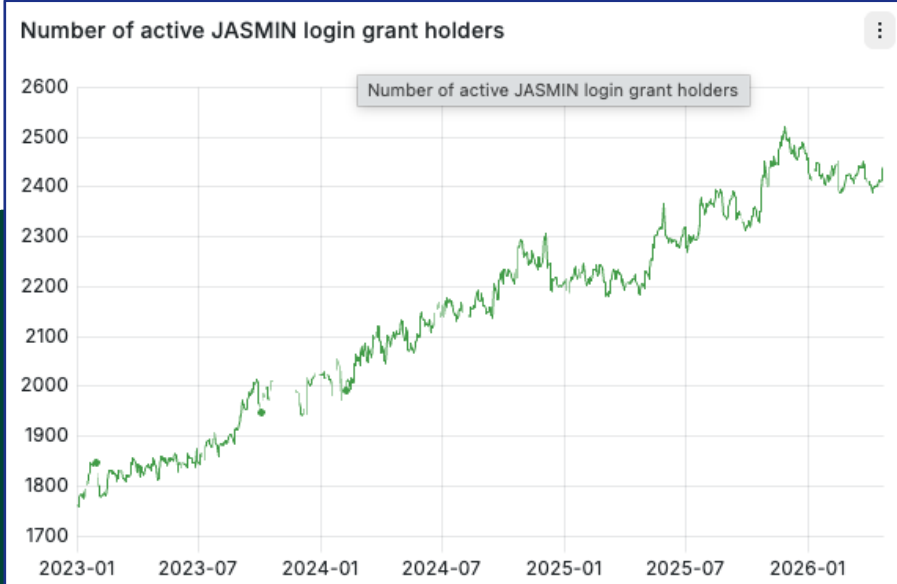


Centre for Environmental
Data Analysis
SCIENCE AND TECHNOLOGY FACILITIES COUNCIL
NATURAL ENVIRONMENT RESEARCH COUNCIL



Services





 **2400 users**
400+ projects

 **~350 kW**
Power consumption

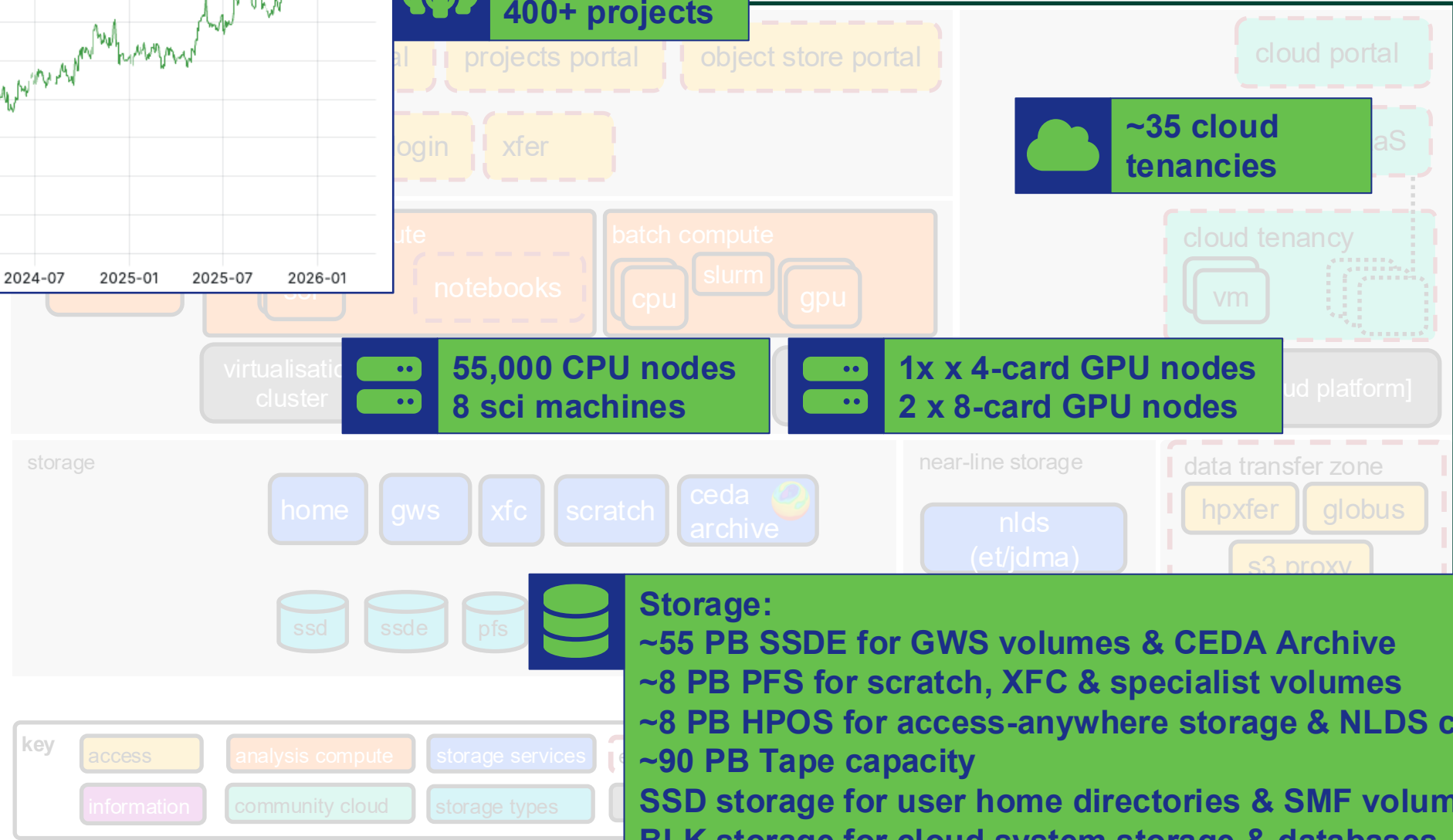
 **~35 cloud**
tenancies

 **55,000 CPU nodes**
8 sci machines

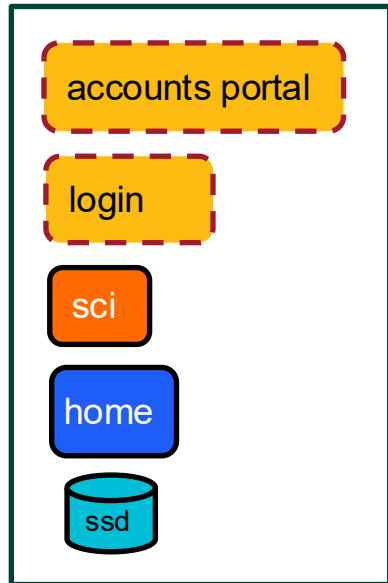
 **1x x 4-card GPU nodes**
2 x 8-card GPU nodes

 **Storage:**
~55 PB SSDE for GWS volumes & CEDA Archive
~8 PB PFS for scratch, XFC & specialist volumes
~8 PB HPOS for access-anywhere storage & NLDS cache
~90 PB Tape capacity
SSD storage for user home directories & SMF volumes
BLK storage for cloud system storage & databases

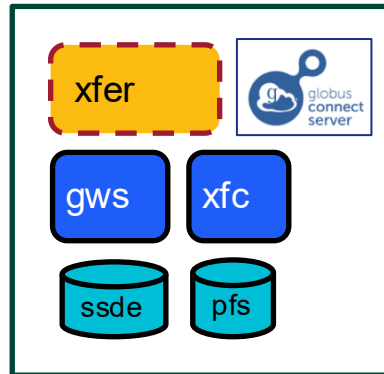
- search
- training
- resources
- dashboard
- status
- discussions



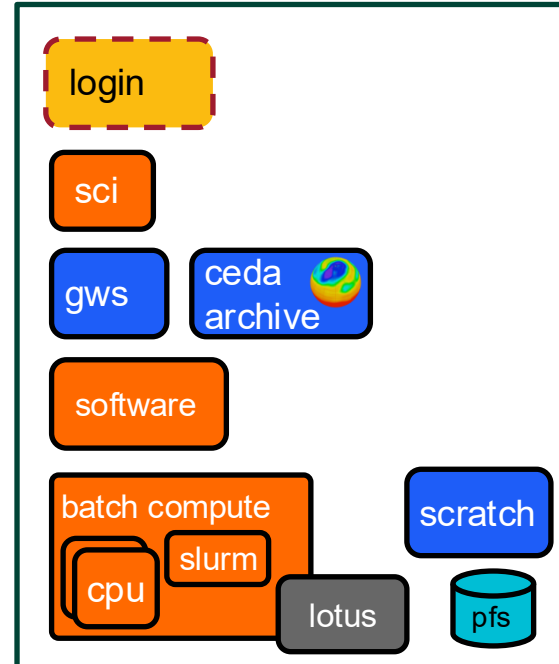
Different ways of using JASMIN



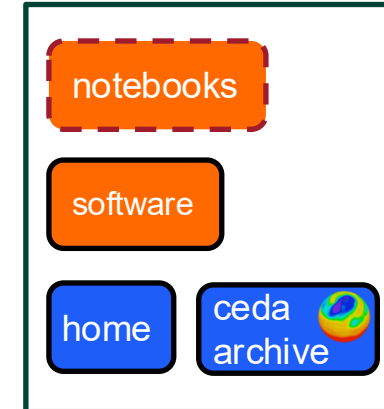
Log in & access home directory.



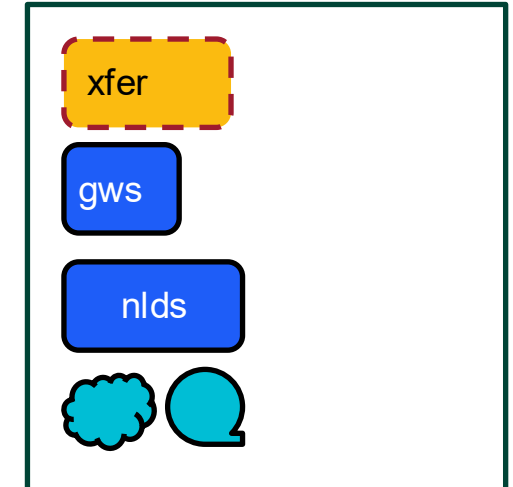
Transfer data, share with collaborators.



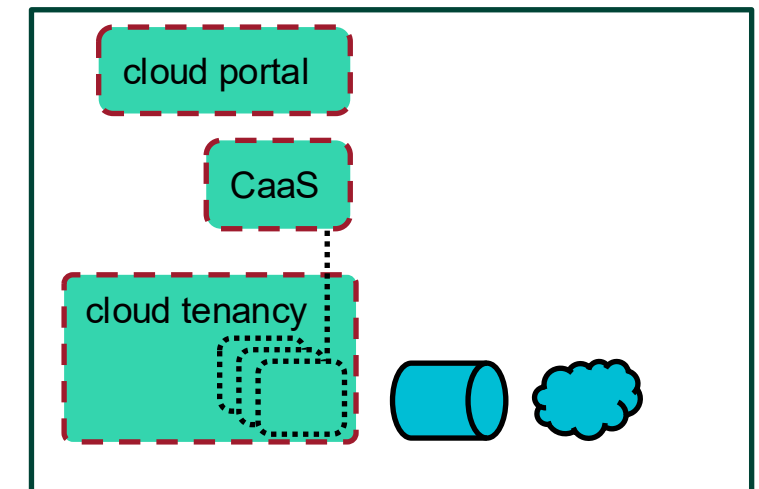
Develop a process on a sci server.
Process at scale on lotus cluster.



Visualise CEDA Archive data in a Jupyter Notebook.



Move data to/from near-line storage.



Develop cloud-native workflows for operational deployment elsewhere.

Resources for projects

JASMIN

getting started >
interactive computing >
batch computing >
software on jasmin >
data transfer >
short term project storage >
long term archive storage >
mass for cloud tenants >
workflow management >
uncategorized >
acknowledging jasmin
approving requests for access
jasmin conditions of use
mobaxterm
processing requests

Home / Docs / Uncategorized / Requesting resources

Requesting resources

Quota Requirement Gws Workspace Tenancy

Share via

Requesting resources

How projects and resources are managed

Resources on JASMIN, such as storage and compute, are allocated to science communities "consortia". Each consortium has a manager: a representative of that science community with its major activities and understands the resource requirements for projects in that domain. Representatives of individual projects should discuss requirements with their Consortium Manager, who is best placed to make decisions about the allocation of JASMIN resources within that consortium. Requirements can be documented using the "JASMIN Resource Management" tool, but need to be approved by a Consortium Manager before being passed to the JASMIN Team for provision.

See here for further details, including a list of current consortia and their managers.

Accessing the JASMIN projects portal

The JASMIN projects portal provides a place to:

projects.jasmin.ac.uk/all_projects

JASMIN My Projects Consortia All Projects

mpritch

Projects List

View of all projects, including filters for tags and consortia. Some columns can be sorted.

Earliest and latest end dates for the project are taken from the requirements in the services of the project. The earliest being the soonest expiry date for a requirement and the latest being the furthest away expiry date for a requirement.

aria All Consortia

Tags	Earliest End Date	Latest End Date	QuoByte SOF	Panasas PFS	Caringo Object Store HPOS	PURE SSD Storage	Tape Storage	BLK storage for VIO7 cloud	RAM for VIO7 cloud	VCPU for VIO7 cloud	ext IP addresses VIO7 cloud	managed IP addresses Cloud	PURE SSDE Storage
ARIA_GIANT	2028-07-04	2028-07-04	20000	—	—	—	—	—	—	—	—	—	—
PROMOTE	2031-03-31	2031-03-31	—	78000	—	—	440000	—	—	—	—	—	—

Useful links

JASMIN
CEDA
Accounts Portal
Cloud Portal
Notebook Service

Contact us

Documentation
Helpdesk
Twitter
YouTube

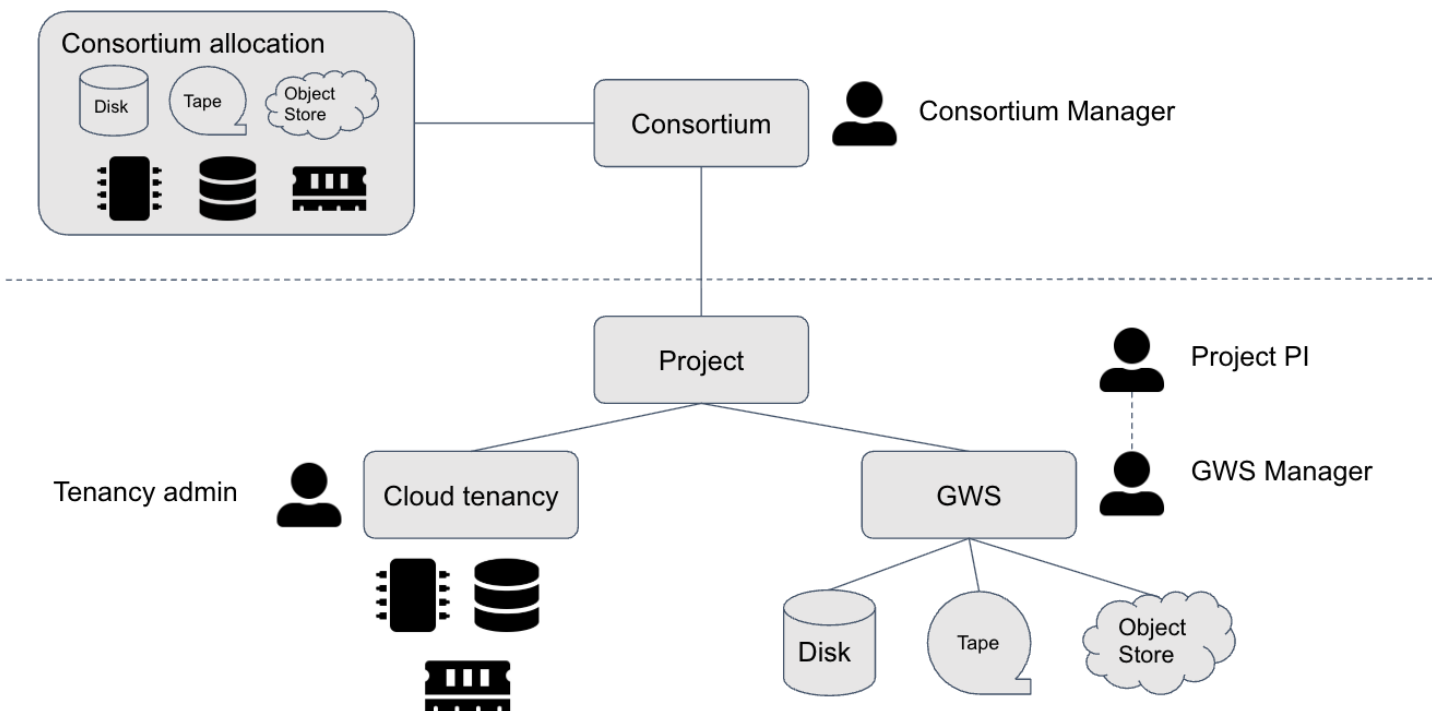
JASMIN is operated by the Science and Technology Facilities Council on behalf of the Natural Environment Research Council.

Science and Technology Facilities Council
Natural Environment Research Council

Help ?

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Community management of resources



JASMIN Consortia

Abbreviation	Name	Manager
atmos	Atmospheric Science	David Hassell, University of Reading
director	Director's cross-cutting activities	Adrian Hines, CEDA
ecohydro	Ecology & Hydrology	Mike Brown, UKCEH
eoclim	Earth Observation & Climate	Federica Moscato & Alison Waterfall, CEDA
genomics	Genomics	TBD
geology	Geology	TBD
npg	National Public Good	Adrian Hines, CEDA
oceanss	Ocean & Shelf Seas	Andrew Coward, NOC
polsci	Polar Science	TBD
semp	Solid Earth & Mineral Physics	John Brodholt, UCL
ukncsp	UK National Climate Science Partnership (was JWRCF)	Bryan Lawrence, University of Reading

The image displays a grid of 10 icons, each representing a different service or resource available on the JASMIN website. The icons are arranged in a 3x3 grid, with the last cell containing a 'Help' button. The icons are as follows:

- Top Row:**
 - About this site:** An information icon (a lowercase 'i' inside a circle).
 - Getting Started:** A play button icon (a right-pointing triangle).
 - Interactive computing:** A computer monitor icon.
- Middle Row:**
 - Batch Computing:** A server rack icon (two horizontal bars with dots).
 - Software on JASMIN:** A code icon (less-than and greater-than signs with a slash).
 - Data transfer:** A double-headed arrow icon.
- Bottom Row:**
 - Long-term archive storage:** A database icon (three stacked cylinders).
 - Help:** A question mark icon inside an orange rounded rectangle.



The screenshot shows a Google Chrome browser window. The top of the window displays several tabs, each with a green icon and the text 'JAS!'. The address bar shows a green icon and the text 'School'. Below the address bar, the page content is mostly green. The text 'Docs' and 'Discussions' is visible, with 'Docs' underlined. There is also a search icon and a toggle switch.

- CEDA News
- Status page
- JASMIN-USERS email list
- JASMIN Help docs
- Discussion fora on GitHub

getting started

interactive computing

access from vscode

creating a virtual environment in the notebooks service

dash gateway

graphical linux desktop access using nx

interactive computing overview

jasmin notebooks service

jasmin notebooks service with gpus

login problems

login servers

project specific servers

sci servers

transfer servers

batch computing

JASMIN Notebooks Service

[Jupyter](#)

Share via [G](#) [F](#) [T](#) [W](#) [S](#) [P](#)

Jupyter Notebooks service on JASMIN

The JASMIN Notebooks Service provides access to [Jupyter Notebooks](#) in a web browser.

What is a Jupyter Notebook?

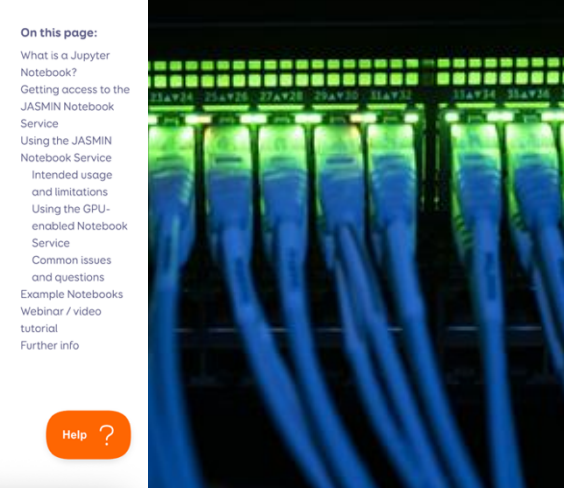
A Jupyter Notebook is an interactive document containing live code and visualisations that can be viewed and modified in a web browser. These documents can be shared, often using GitHub, and many projects distribute example code as Jupyter notebooks. Users interact with their notebooks using the open-source Jupyter Notebook server application.




On this page:

- What is a Jupyter Notebook?
- Getting access to the JASMIN Notebook Service
- Using the JASMIN Notebook Service
- Intended usage and limitations
- Using the GPU-enabled Notebook Service
- Common issues and questions
- Example Notebooks
- Webinar / video tutorial
- Further info

Help ?



Training

Training events

- JASMIN Workshop
- Hackathons
- Tailored training
- Support for other events



Training accounts

Preconfigured with required
access
Used for our training events
Can be “borrowed” for yours

JASMIN

JASMIN training accounts

In this article:

- [Overview](#)
- [For event organisers](#)
- [For event attendees](#)

Overview

What are training accounts?

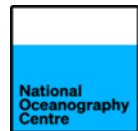
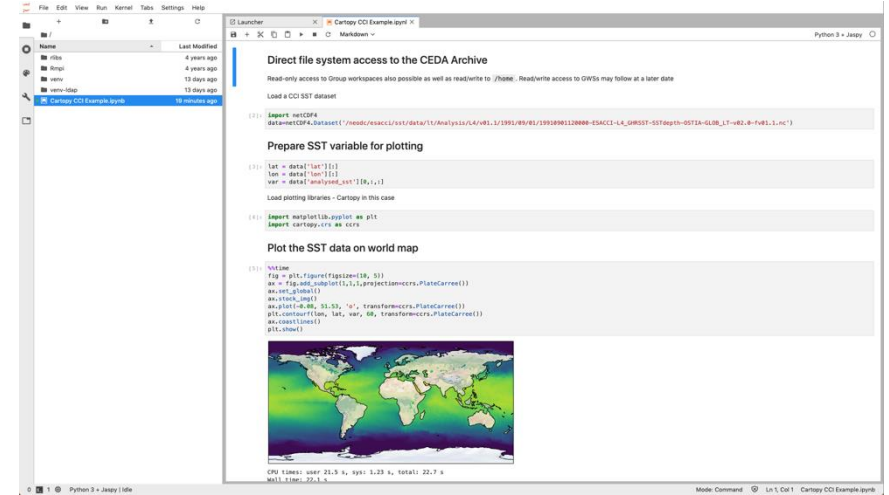
JASMIN training accounts are TEMPORARY accounts that provide a person with access to JASMIN for a specific event (such as a training workshop) organised in advance.

The benefits of using these accounts are:

- permissions/services are consistent across all training accounts

Current priorities (April 2026)

- Complete storage replacement
- ORCHID GPU cluster: develop case for replacement
- Environmental sustainability
- Exploring resilience options
- Increase support to Hackathons & 3rd-party training
- Developing closer links with other DRIs
- Working with other communities
- Further integration into UK Digital Research Infrastructure

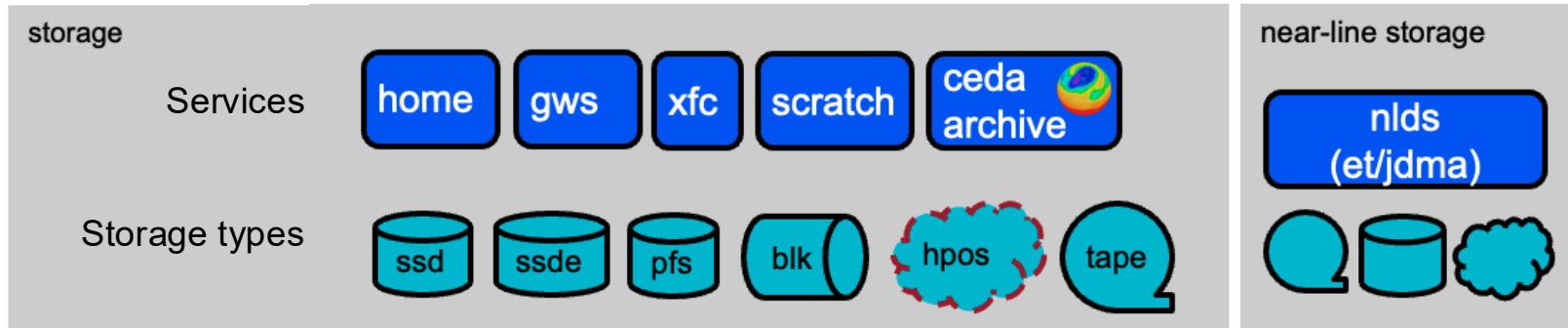


In more detail...

- Storage
- Compute
- Software
- Cloud
- Data transfer
- Your JASMIN account
- Expectations

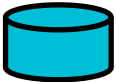


Storage



- Mix of storage types to support several services

Storage features

	Features				
	Disk (file system)	✓	✓	✓	✓
	Parallel Write	 OK except parallel netCDF	 OK except parallel netCDF	✓	✓
	Small files (<8Mb)	✓	✓		
	Other				Phasing out during 2026
	Near-line	NLDS: JASMIN Group workspace data Also used for CEDA Archive backup & staging			
	Object Store	Interface like Amazon S3. Disk-based, but simpler with no file system: just buckets & objects. Access from anywhere. Tape-like storage for GWS & cloud-based workflows.			
	Block	For cloud tenancy storage and other internal uses (virtualisation, databases)			



Storage deployments

		Path(s)	Mounts						Backup	Other
			login	(hp)xfer	g	sci	nb	lotus orchid		
home	ssd	/home/users/<username>	✓	✓	✓	✓	✓	✓	✓	100GB quota
gws	ssde pfs ssd	/gws/ssde/j25a/<gwsname> /gws/ssde/j25b/<gwsname> /gws/pw/j07/<gwsname> /gws/smf/j07/<gwsname> /gws/nopw/j04/<gwsname>		✓	✓	✓	✓	✓	✗	One or more volumes per project. Sized to project requirements (typically 1Tb – 1Pb) “SMF” vols on SSD were for small files, may not be needed now most GWSs moving to SSDE.
scratch	pfs ssd	/work/scratch-pw4 /work/scratch-pw5 /work/scratch-nopw2			✓	✓	✓	✓	✗	For use only as part of LOTUS/ORCHID cluster jobs. Auto-deletion of files older than 28 days.
xfc	pfs	/work/xfc/volN/user_cache/<username>		✓	✓	✓	✓	✓	✗	For transient use in transfer tasks. User clean-up expected!
ceda archive	pfs ssde	/badc /neodc ..dc		✓	✓	✓	✓	✓	✓	Read-only access throughout JASMIN, if CEDA account linked. No need to copy archive data: process in-place.

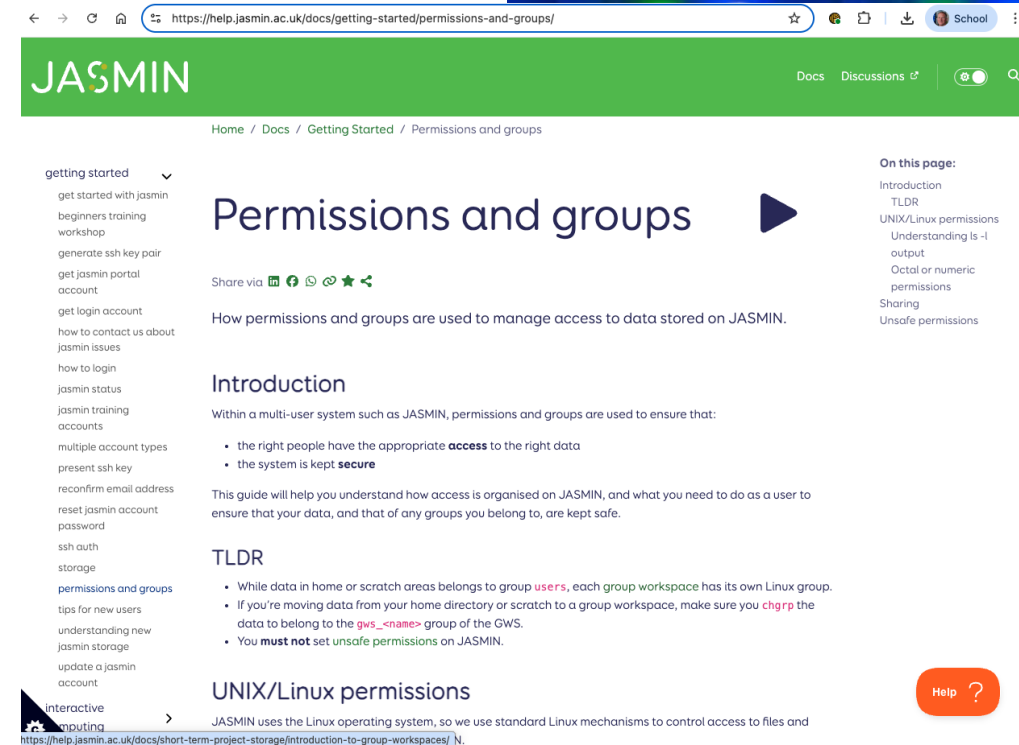
Other storage resouces

- Object storage (S3)
 - Simpler than file system: buckets + put/get
 - Access-anywhere
- Near-Line Data Store (NLDS)
 - Command-line interface + S3 cache + Tape storage + catalogue
 - Interface to put / get data from local disk, manage holdings
 - Encouraged vs storing all data online – lower energy



Group Workspaces

- GWS Manager controls membership
 - You apply, manager can approve/deny
 - GWS Manager can also just “grant”
 - GWS has own unix group
 - See “permissions and groups”
- GWSs are NOT BACKED UP
- For science data only
- Not permanent
- Consortium manager



The screenshot shows the JASMIN website's 'Permissions and groups' page. The page has a green header with the JASMIN logo and navigation links for Docs, Discussions, and a search icon. A sidebar on the left lists various 'getting started' topics, with 'permissions and groups' highlighted. The main content area is titled 'Permissions and groups' and includes a play button icon. Below the title, there's a 'Share via' section with social media icons. The text explains that permissions and groups are used to manage access to data stored on JASMIN. It includes an 'Introduction' section stating that within a multi-user system, permissions and groups ensure that the right people have appropriate access and the system is kept secure. A 'TLDR' section follows, summarizing that data belongs to a group (users), each workspace has its own Linux group, and users must not set unsafe permissions. The page also features a 'UNIX/Linux permissions' section and a 'Help' button in the bottom right corner.



Compute

- Sci servers
 - interactive processing
 - Compilation
 - Testing, estimation of resources needed
 - Job submission to LOTUS, monitoring
 - Common software stack installed
 - Not for huge processing!
- LOTUS batch cluster
 - for tasks which don't need to be run interactively
 - Job scheduler: **Slurm**, applies fair-share system
 - Same software as interactive compute
 - Same storage resources
 - Enables compilation on specific hardware

Sci machines for general use:

5 virtual machines `sci-vm-0[1-5].jasmin.ac.uk`

2 physical machines `sci-ph-0[1-3].jasmin.ac.uk`

See **Exercise 2**

- Over 55,000 cores (277 nodes)
- AMD Architecture
- Scheduler partitions (queues) with different attributes (QoS)
- MPI parallel processing possible

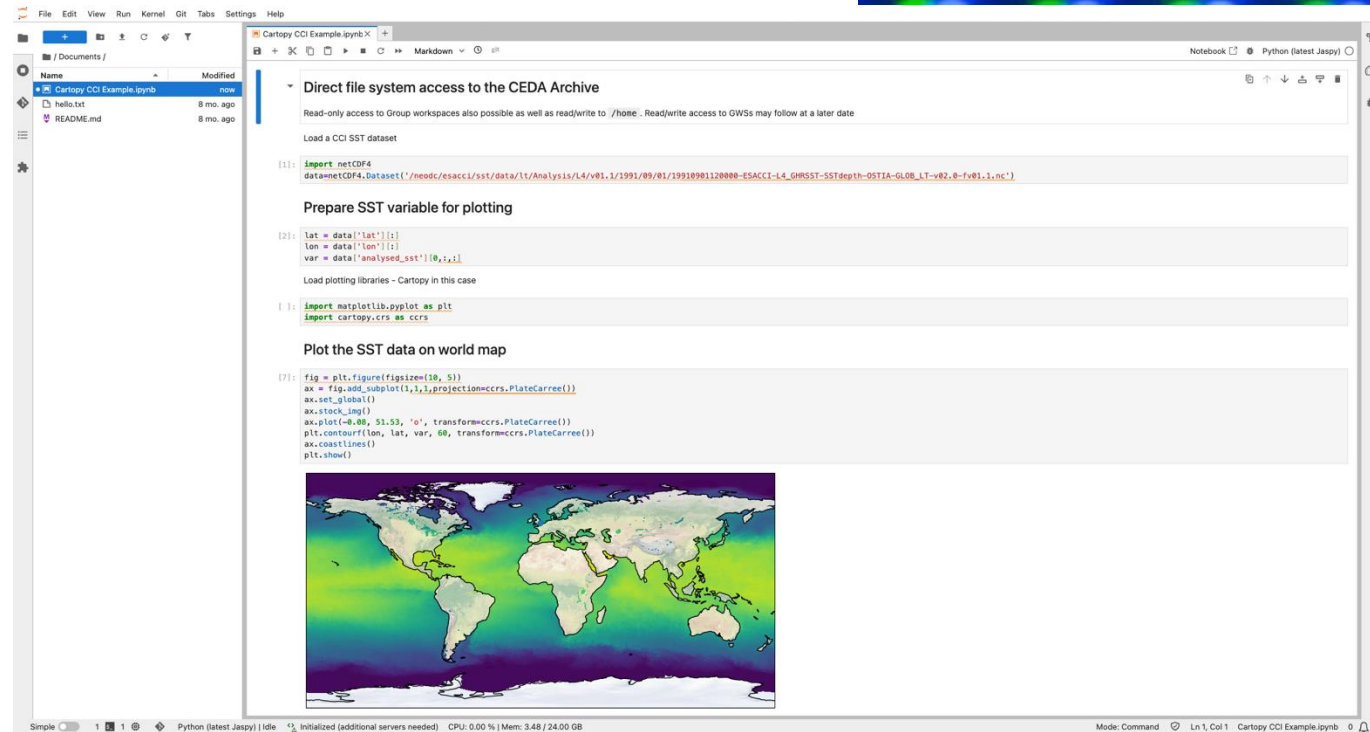
See **Exercise 5**

Training exercises:

<https://github.com/cedadev/jasmin-workshop>

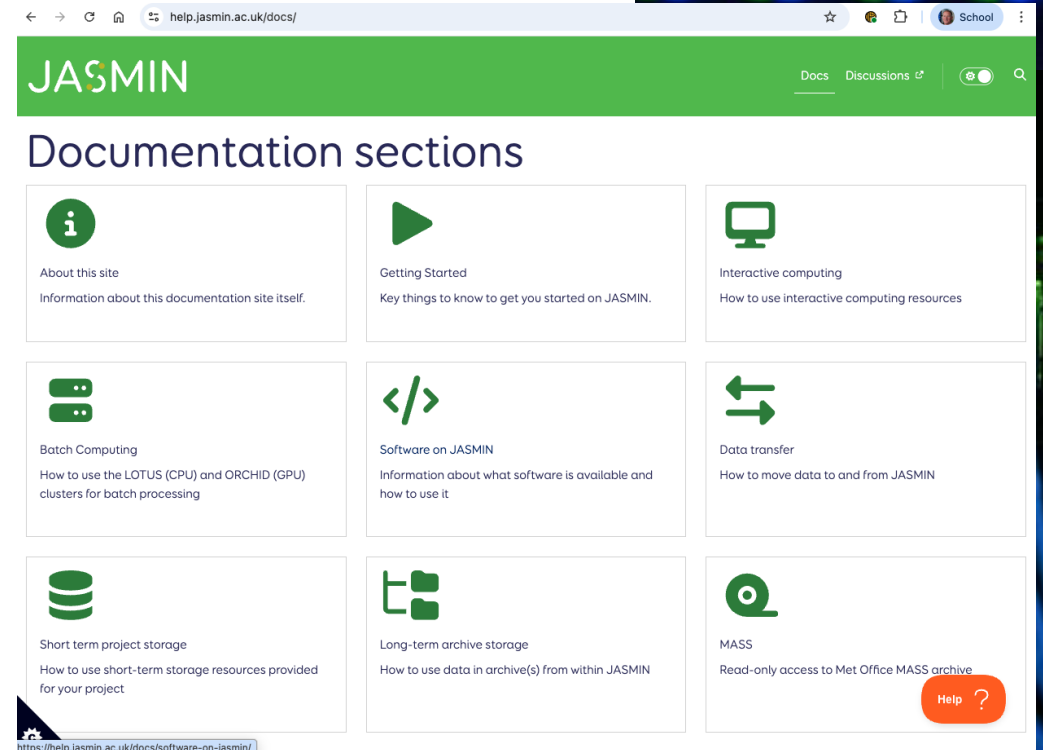
Notebooks service

- Use JASMIN interactively
 - Jupyter Notebooks: live code & output in web interface
 - Great for quick visualisations
 - Now possible to use GPU too
- See Exercise 9



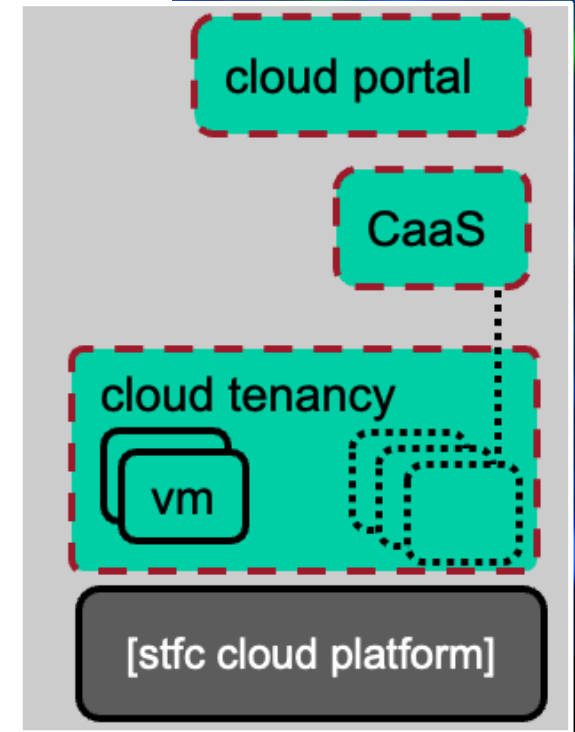
Software

- Analysis software common to batch & interactive compute
 - Stack developed with user community
 - `jaspy`, `jasr`, `jasmin-sci`
 - Additional tools for compiling and building software
 - Restricted software
 - Data movement software
-
- Same software available across
 - `sci` / `notebooks` / `lotus`



Cloud

- Separate part of infrastructure
 - STFC Cloud Platform: Openstack
- Create bespoke compute setup
 - VMs
 - Cluster-as-a-Service : use “building blocks”
 - Managed via “Azimuth” cloud portal
 - <https://github.com/azimuth-cloud/azimuth>



Data transfer

- Dedicated services

xfer

- **xfer servers** (VMs)
 - small / everyday stuff
 - SSH-based transfer tools: scp / sftp / rsync

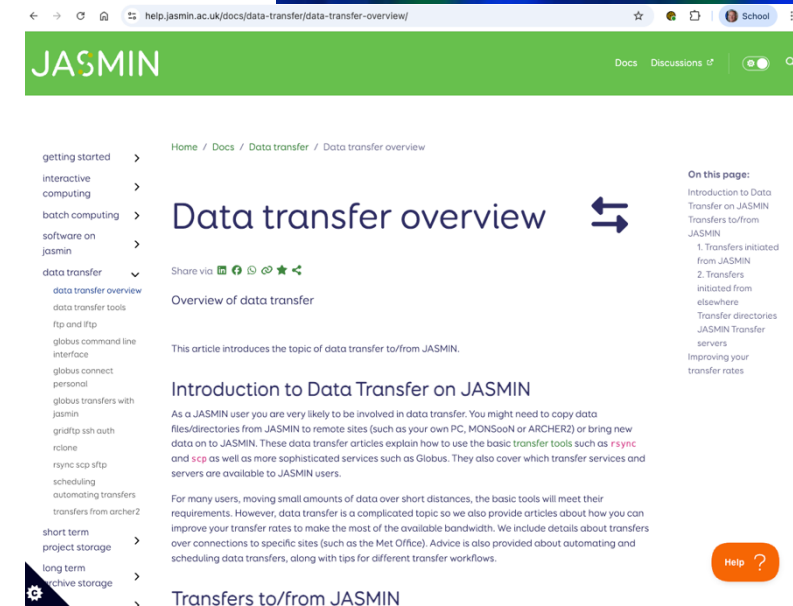
hpxfer

- **hpxfer servers** (physical servers, higher performance)
 - larger datasets / longer distances to JASMIN from elsewhere

- globus

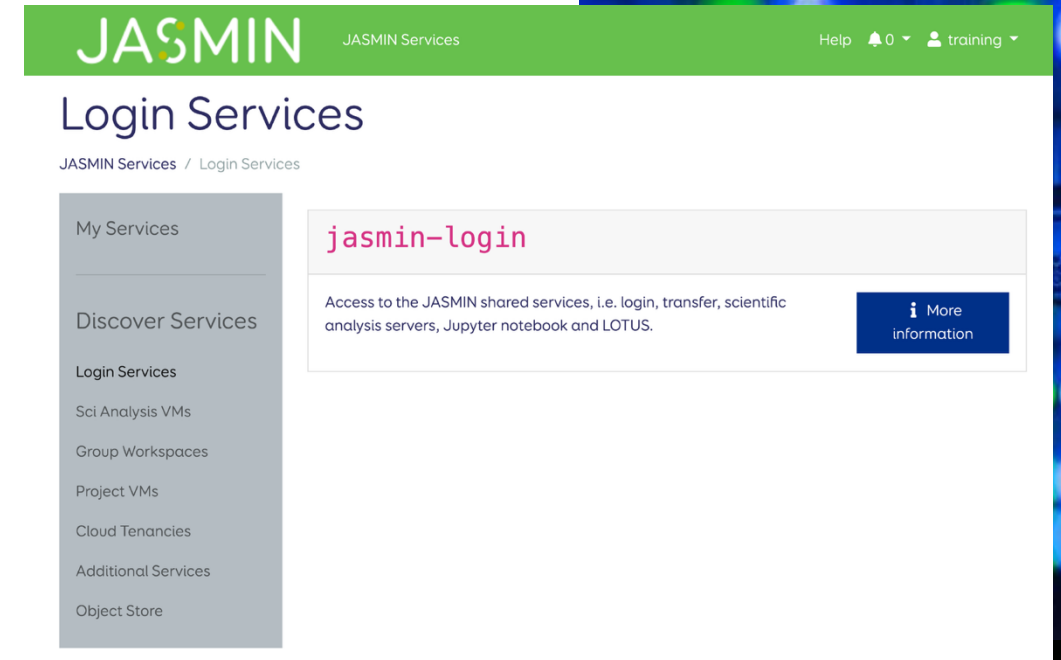


- best performance, convenience, reliability
- set up regular tasks
- asynchronous, managed, auto-retry, ...
- web / CLI / SDK interfaces



Your JASMIN account

- Each user has a JASMIN account
 - Profile to attach privileges for JASMIN services
 - **jasmin-login** = needed for SSH login access & Notebooks service
 - GWS membership(s)
 - Apply for services at JASMIN Accounts Portal
 - GWS membership is approved by GWS manager (not CEDA/JASMIN team)
 - Different to your CEDA account, if you have one
 - Needs to be linked for CEDA Archive access



The screenshot shows the JASMIN web interface. At the top is a green header with the JASMIN logo, 'JASMIN Services', and navigation links for 'Help', '0', and 'training'. Below the header, the page title is 'Login Services'. A breadcrumb trail reads 'JASMIN Services / Login Services'. On the left is a grey sidebar menu with links: 'My Services', 'Discover Services', 'Login Services', 'Sci Analysis VMs', 'Group Workspaces', 'Project VMs', 'Cloud Tenancies', 'Additional Services', and 'Object Store'. The main content area features a box for 'jasmin-login' with a description: 'Access to the JASMIN shared services, i.e. login, transfer, scientific analysis servers, Jupyter notebook and LOTUS.' To the right of this text is a blue button labeled 'More information'.



Your JASMIN account

- Each user has a JASMIN account
 - Profile to attach privileges for JASMIN services
 - **jasmin-login** = needed for SSH login access & Notebooks service
 - GWS membership(s)
 - Apply for services at JASMIN Accounts Portal
 - GWS membership is approved by GWS manager (not CEDA/JASMIN team)
 - Different to your CEDA account, if you have one
 - Needs to be linked for CEDA Archive access

If you register as a JASMIN user yourself

- Accept T&Cs
- Please read reminder emails

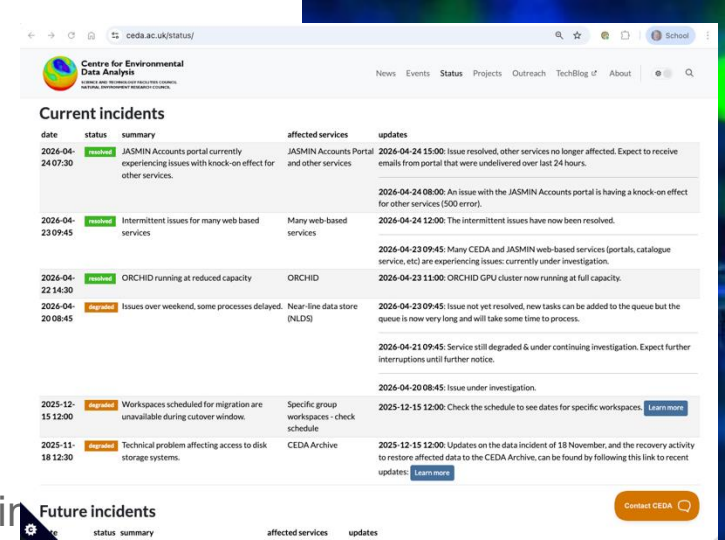
Once a year, re-confirm your account

- Each role: expiry reminders 2 months / 2 weeks / 2 days before (renew in advance!)



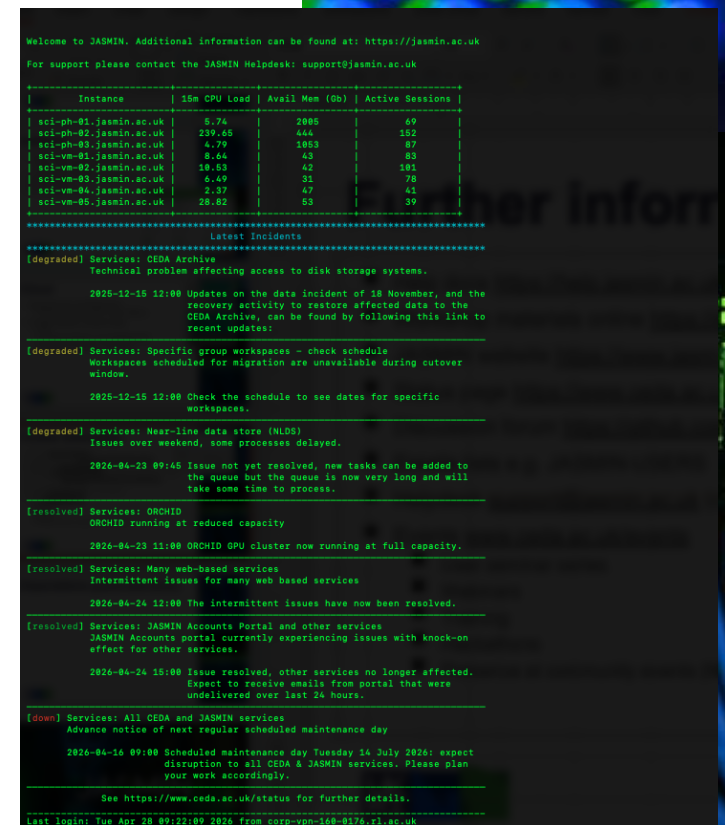
Further information

- Help docs <https://help.jasmin.ac.uk>
- Workshop materials online <https://github.com/cedadev/jasmin-workshop>
- JASMIN website <https://www.jasmin.ac.uk>
- Status page <https://www.ceda.ac.uk/status> & message-of-the-day (MOTD) in terminal
- Discussion forum <https://github.com/cedadev/jasmin-community/discussions>
- Emails lists e.g. JASMIN-USERS
- Helpdesk support@jasmin.ac.uk (or use Help Beacon)
- Events www.ceda.ac.uk/events
 - User seminar series
 - Webinars
 - Training
 - Hackathons
 - Presence at community events (NCAS, NCEO, NERC-DG)



The screenshot shows the CEDA status page with a table of current incidents. The table has columns for date, status, summary, affected services, and updates. Incidents include JASMIN Accounts portal issues, intermittent web-based service issues, ORCHID running at reduced capacity, and workspace migration issues.

date	status	summary	affected services	updates
2026-04-24 07:30	resolved	JASMIN Accounts portal currently experiencing issues with knock-on effect for other services.	JASMIN Accounts Portal and other services	2026-04-24 15:00: Issue resolved, other services no longer affected. Expect to receive emails from portal that were undelivered over last 24 hours.
2026-04-23 09:45	resolved	Intermittent issues for many web-based services	Many web-based services	2026-04-24 08:00: An issue with the JASMIN Accounts portal is having a knock-on effect for other services (500 error). 2026-04-24 12:00: The intermittent issues have now been resolved.
2026-04-22 14:30	resolved	ORCHID running at reduced capacity	ORCHID	2026-04-23 09:45: Many CEDA and JASMIN web-based services (portals, catalogue service, etc) are experiencing issues: currently under investigation. 2026-04-23 11:00: ORCHID GPU cluster now running at full capacity.
2026-04-20 08:45	resolved	Issues over weekend, some processes delayed.	Near-line data store (NLDS)	2026-04-23 09:45: Issue not yet resolved, new tasks can be added to the queue but the queue is now very long and will take some time to process. 2026-04-21 09:45: Service still degraded & under continuing investigation. Expect further interruptions until further notice. 2026-04-20 08:45: Issue under investigation.
2025-12-15 12:00	resolved	Workspaces scheduled for migration are unavailable during cutover window.	Specific group workspaces - check schedule	2025-12-15 12:00: Check the schedule to see dates for specific workspaces. Learn more
2025-11-18 12:30	resolved	Technical problem affecting access to disk storage systems.	CEDA Archive	2025-12-15 12:00: Updates on the data incident of 18 November, and the recovery activity to restore affected data to the CEDA Archive, can be found by following this link to recent updates: Learn more



The screenshot shows the JASMIN terminal MOTD. It includes a welcome message, support contact information, a table of instance statistics, and a list of latest incidents with their status and affected services.

```

Welcome to JASMIN. Additional information can be found at: https://jasmin.ac.uk
For support please contact the JASMIN Helpdesk: support@jasmin.ac.uk

+-----+
| Instance | 15m CPU Load | Avail Mem (Gb) | Active Sessions |
+-----+
| sci-ph-01.jasmin.ac.uk | 5.74 | 2005 | 49 |
| sci-ph-02.jasmin.ac.uk | 239.65 | 444 | 152 |
| sci-ph-03.jasmin.ac.uk | 4.79 | 1053 | 87 |
| sci-vm-01.jasmin.ac.uk | 6.64 | 63 | 83 |
| sci-vm-02.jasmin.ac.uk | 10.53 | 42 | 101 |
| sci-vm-03.jasmin.ac.uk | 6.49 | 31 | 78 |
| sci-vm-04.jasmin.ac.uk | 2.37 | 47 | 41 |
| sci-vm-05.jasmin.ac.uk | 26.82 | 53 | 39 |
+-----+

*****
Latest Incidents
*****
(degraded) Services: CEDA Archive
Technical problem affecting access to disk storage systems.

2025-12-15 12:00 Updates on the data incident of 18 November, and the
recovery activity to restore affected data to the
CEDA Archive, can be found by following this link to
recent updates:

(degraded) Services: Specific group workspaces - check schedule
Workspaces scheduled for migration are unavailable during cutover
window.

2025-12-15 12:00 Check the schedule to see dates for specific
workspaces.

(degraded) Services: Near-line data store (NLDS)
Issues over weekend, some processes delayed.

2026-04-23 09:45 Issue not yet resolved, new tasks can be added to
the queue but the queue is now very long and will
take some time to process.

(resolved) Services: ORCHID
ORCHID running at reduced capacity

2026-04-23 11:00 ORCHID GPU cluster now running at full capacity.

(resolved) Services: Many web-based services
Intermittent issues for many web-based services

2026-04-24 12:00 The intermittent issues have now been resolved.

(resolved) Services: JASMIN Accounts Portal and other services
JASMIN Accounts portal currently experiencing issues with knock-on
effect for other services.

2026-04-24 15:00 Issue resolved, other services no longer affected.
Expect to receive emails from portal that were
undelivered over last 24 hours.

(down) Services: All CEDA and JASMIN services
Advance notice of next regular scheduled maintenance day

2026-04-16 09:00 Scheduled maintenance day Tuesday 16 July 2026: expect
disruption to all CEDA & JASMIN services. Please plan
your work accordingly.

See https://www.ceda.ac.uk/status for further details.

Last login: Tue Apr 10 09:12:09 2026 from corp-vm-168-0176.rl.ac.uk
```

Expectations (yours and ours!)

- **JASMIN is a research infrastructure**
 - At cutting edge of data-intensive environmental informatics
 - Not a 24x7x365 enterprise
 - Small team, no out-of-hours cover
- **More flexible than a supercomputer, but...**
 - Relies on you being sensible
 - You need to know what you're doing
 - Test stuff before scaling up!
 - Build failure-tolerance into your workflows
- **Please acknowledge JASMIN**
 - <https://help.jasmin.ac.uk/docs/uncategorized/acknowledging-jasmin/>
 - Tell your funding body what you've done with JASMIN
 - ...And tell us! (impact stories)
 - Helps ensure continued funding
 - Makes the case for improving the service further



JASMIN



Science and
Technology
Facilities Council



Natural
Environment
Research Council